

GovRed-Health: Benchmarking Authorization Drift Between Governed Disclosure and Persistent Commit

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Abstract—Security benchmarks for LLM agents now cover prompt injection, tool misuse, memory attacks, and generic commit-time authorization. Persistent clinical AI introduces a narrower yet critical failure mode: a task-bounded governed health disclosure may be fully authorized when an LLM reads it, but the underlying patient state, dynamic consent, governance policy, role, purpose, cache entry, or bitemporal validity can change before the model’s derived proposal is committed to durable storage. We formalize this hazard as *bitemporal authorization drift across the disclosure-to-commit interval*. GovRed-Health is a real-boundary adversarial benchmark designed to evaluate this failure mode. We incorporate the Santos-Grueiro four-boundary commit-time authorization framework (Freshness, Causal Precedence, Effect Scoping, and Admissibility) into matched logical schedules that execute through authentic HTTP/API gateways, authorization filters, cache/index stores, PostgreSQL databases, and audit ledgers. Furthermore, we evaluate how Task-Bounded Health State Snapshot (THSS) compilation achieves an 87.4% prompt token reduction (mean 412 vs. 3,280 tokens) and KV-cache prefix optimization, eliminating cache thrashing and latency overhead while guaranteeing zero over-disclosure of protected health information. In a sealed isolated run (revision 5b2c0dbf17e2, run 2026-08-17-rivf-final-003) across 270 logical schedules per arm (210 executable primary schedules), the prespecified non-safe composite was 57.1% for Unbound, 42.9% for State-Version-Only, 42.9% for Snapshot-Bound-State-Only, and 14.3% for GLHS-Strict ($p = 1.62 \times 10^{-27}$ vs. Unbound). All 30 Strict residuals were concurrent stale-state schedules with indeterminate linearization; the tested serial authorization-drift families produced zero confirmed invalid Strict commits, and prohibited disclosure was 0/270 in every arm.

Index Terms—health AI, bitemporal authorization drift, Santos-Grueiro 4-boundary, commit-time authorization, KV-cache optimization, token reduction, TOCTOU, dynamic consent, persistent agents

I. INTRODUCTION

Security research for large language model (LLM) agents has advanced from descriptive jailbreak demonstrations toward executable, reproducible benchmarks. AgentDojo evaluates prompt injection against tool-using agents [1]; Agent Security Bench covers multi-vector attacks and defenses [2]; MPIB investigates direct and RAG-mediated prompt injection in clinical tasks [3]; and SecureClaw

introduces dual-boundary enforcement for sensitive reads and external effects [4].

A separate and crucial line of systems research addresses stateful and transactional agent execution. Santos-Grueiro formalized commit-time authorization, showing that authority acquired during an agent trajectory can become stale before durable effects take place [5]. Stateful governance research defines policy-state serializability for concurrent agents [6], while GateMem benchmarks access control and forgetting in multi-principal shared memory [7]. Consequently, generic stale authorization, memory access control, and commit-time checks are established problems.

Persistent clinical AI, however, introduces a domain-specific cross-operation hazard centered on the *governed health disclosure actually consumed by the model*. In longitudinal electronic health records (EHR), clinical facts possess distinct bitemporal coordinates [8], [9]: valid time (t_v , when a clinical assertion is true in the patient’s physiology) and transaction/knowledge time (t_k , when it is recorded in the system). An AI system may construct a Task-Bounded Health State Snapshot (THSS) for a specific patient, actor, role, purpose, consent state, and due window at knowledge time $t_k^{(1)}$, but receive the model-generated treatment proposal only at $t_k^{(2)} > t_k^{(1)}$. During this read-to-write interval $[t_k^{(1)}, t_k^{(2)}]$, dynamic patient consent may be revoked, hospital policies updated, patient state modified by concurrent clinicians, or retrospective corrections ingested into the bitemporal ledger. A proposal committed under stale authority creates a Time-of-Check to Time-of-Use (TOCTOU) vulnerability that can lead to catastrophic medical errors, privacy violations, or regulatory non-compliance.

Moreover, stateful clinical agents face an efficiency barrier. Feeding complete, unminimized longitudinal patient histories to LLMs incurs massive prompt overhead (averaging 3,280 tokens per request), degrading inference throughput and causing key-value (KV) cache thrashing and memory eviction under concurrent workloads. Task-bounded context minimization addresses this by distilling minimal necessary clinical evidence into a canonically ordered, Merkle-hashed prefix, achieving an 87.4% prompt token reduction and enabling KV-cache prefix sharing across reasoning stages.

GovRed-Health turns the bitemporal disclosure-to-

TABLE I
CROSS-BENCHMARK CAPABILITY COMPARISON: THREAT DIMENSIONS
IN STATEFUL CLINICAL AGENT SECURITY.

Benchmark	Prompt Attack	Tool Access	Bitemporal TOCTOU
AgentDojo [1]	✓	✓	×
MPIB [3]	✓	×	×
CommitGuard [5]	×	✓	✓
Provenact [6]	×	✓	✓
GateMem [7]	×	✓	×
GovRed-Health (Ours)	✓	✓	✓

commit interval into an empirical benchmark. Each matched schedule follows an invariant sequence: *governed disclosure* $\rightarrow$ *controlled bitemporal/governance mutation* $\rightarrow$ *persistent commit attempt*, executed across real application, authorization, database, and audit boundaries. We map all attack vectors against the Santos-Grueiro four-boundary commit-time authorization model [5] (Freshness, Causal Precedence, Effect Scoping, and Admissibility), providing the first rigorous quantification of bitemporal authorization drift in clinical AI systems.

II. RELATED WORK AND NOVELTY BOUNDARY

Prompt- and tool-centric benchmarks. AgentDojo, ASB, and MPIB provide adversarial testbeds for injection and tool hijacking [1], [2], [3]. GovRed-Health treats prompt-level stress tests strictly as a secondary category; prompt injection is not our primary novelty claim.

Dual-boundary architectures. SecureClaw isolates sensitive data reads from external side effects [4]. GovRed-Health builds upon dual-boundary concepts but specifically evaluates the temporal validity of the disclosure object across real persistence boundaries.

Commit-time authorization and bitemporal systems. Santos-Grueiro established commit-time authorization for LLM agents, formalizing the requirement that authority witnesses must remain valid at durability [5]. Stateful governance formalizes policy-state serializability [6], GateMem evaluates multi-principal memory [7], and TOKI formalizes bitemporal operator algebras for agent memory [11].

Novelty boundary. GovRed-Health does not claim bitemporal databases, optimistic concurrency control (OCC), provenance tracking, or generic commit-time authorization as new. The specific contribution is a **health-specific benchmark whose experimental unit couples the model-consumed governed bitemporal disclosure, an intervening governance/state mutation, the persistent commit attempt under the Santos-Grueiro 4-boundary framework, and end-to-end audit reconstruction.**

III. BITEMPORAL AUTHORIZATION DRIFT MODELING

A. Mathematical Formulation

Let clinical evidence in a longitudinal health record be represented bitemporally as $e = (id, u, val, \tau_v, \tau_k, prov)$,

where u is the patient subject, val is the clinical payload (e.g., lab observation, medication order), $\tau_v = [t_{v,start}, t_{v,end}]$ is the valid time interval, $\tau_k = [t_{k,known}, \infty)$ is the transaction/knowledge time interval, and $prov$ is the cryptographic provenance tuple.

× A Task-Bounded Health State Snapshot (THSS), denoted $\mathcal{H}_k^\times$ is compiled by the API Gateway at knowledge time $t_k^{(1)}$ for a specific task τ :

$$\mathcal{H} = (u, a, r, p, \tau_v, t_k^{(1)}, \vec{v}_s, v_g, d_H, e_H, E_H), \quad (1)$$

where a is the actor, r is the active role, p is the clinical purpose of use, $\vec{v}_s \in \mathbb{N}^k$ is the base entity-state version vector, v_g is the governance version (encoding active consent Σ and policy Π), e_H is the cryptographic expiry timestamp, $E_H = \{e_1, \dots, e_m\}$ is the minimal admitted evidence subset, and d_H is the canonical Merkle root digest:

$$d_H = \text{Merkle} \left(E_H \parallel u \parallel a \parallel r \parallel p \parallel \tau_v \parallel t_k^{(1)} \parallel \vec{v}_s \parallel \Sigma \parallel \Pi \right). \quad (2)$$

An LLM reasons over $\mathcal{H}$ and produces a persistent proposal $P = (u_P, \Delta_S, \tau_{v,P}, \mathcal{W}, id_H, d_H)$ submitted at transaction time $t_k^{(2)} > t_k^{(1)}$, where Δ_S is the proposed state delta, $\tau_{v,P}$ is the proposed clinical valid time, and $\mathcal{W}$ is the declared write entity set.

Definition 1 (Bitemporal Authorization Drift). An authorization drift condition $\Delta_{\text{drift}}(\mathcal{H}, P, t_k^{(2)})$ occurs across the interval $[t_k^{(1)}, t_k^{(2)}]$ if and only if:

$$\begin{aligned} \Delta_{\text{drift}} \iff & \left(\exists e \in E_H : \text{status}(e, t_k^{(2)}) \neq \text{status}(e, t_k^{(1)}) \right) \vee \\ & \left(\Sigma(t_k^{(2)}) \neq \Sigma(t_k^{(1)}) \right) \vee \left(\Pi(t_k^{(2)}) \neq \Pi(t_k^{(1)}) \right) \vee \\ & \left(\vec{V}_{\text{db}}(t_k^{(2)}) \neq \vec{v}_s \right) \vee \left(t_k^{(2)} > e_H \right) \vee (u_P \neq u), \end{aligned} \quad (3)$$

where $\vec{V}_{\text{db}}(t_k^{(2)})$ is the current database entity version vector at commit time. Under authorization drift, any persistent commit of P without revalidation constitutes a TOCTOU governance violation.

B. Santos-Grueiro 4-Boundary Commit-Time Authorization

GovRed-Health structures write-admission validation across the four formal commit-time boundaries defined by Santos-Grueiro [5]:

- 1) **Freshness Boundary ($\mathcal{B}_{\text{fresh}}$):** Validates temporal currency $\tau_v \cap \tau_k \neq \emptyset$ and verifies that the snapshot lease has not expired: $t_k^{(2)} \leq e_H$.
- 2) **Causal Precedence Boundary ($\mathcal{B}_{\text{causal}}$):** Verifies optimistic concurrency control invariants $\vec{V}_{\text{db}}(\mathcal{W}, t_k^{(2)}) == \vec{v}_s(\mathcal{W})$, ensuring the causal base state has not been mutated by concurrent transactions.
- 3) **Effect Scoping Boundary ($\mathcal{B}_{\text{scope}}$):** Restricts durable mutations strictly to declared entity partitions: $\mathcal{W} \subseteq \text{Scope}(\mathcal{H}) \wedge (u_P == u)$, preventing cross-patient or collateral data corruption.

TABLE II
CORE AUTHORIZATION-DRIFT SCHEDULES AND 4-BOUNDARY MAPPING.

Schedule Family	Drift Invariant Evaluated	Boundary
Disclosure → Consent Revoke → Commit	Revoked consent blocks persistent writeback.	$\mathcal{B}_{\text{admiss}}$
Disclosure → Policy Update → Commit	Policy drift is revalidated at commit time.	$\mathcal{B}_{\text{admiss}}$
Disclosure → Role/Purpose Change	Active role/purpose cannot silently rebind.	$\mathcal{B}_{\text{admiss}}$
Disclosure → Cross-Subject Replay	Snapshot/proposal cannot migrate across subjects.	$\mathcal{B}_{\text{scope}}$
Disclosure → State Update → Stale Commit	Stale base state cannot overwrite newer state.	$\mathcal{B}_{\text{causal}}$
Disclosure → Revoke → Cache Reuse	Cache cannot license revoked authority.	$\mathcal{B}_{\text{fresh}}$
Snapshot → Expiry/Digest Mutation	Tampered or expired snapshots are rejected.	$\mathcal{B}_{\text{fresh}}$
Concurrent Writer Proposal Commit	State outcome matches first-linearized writer.	$\mathcal{B}_{\text{causal}}$
Audit Reconstruction	Disclosure, mutation, and commit are auditable.	$\mathcal{B}_{\text{all}}$
Secondary Prompt/Bypass Stress	Text injections cannot bypass 4-boundary gate.	$\mathcal{B}_{\text{admiss}}$

- 4) **Admissibility Boundary ($\mathcal{B}_{\text{admiss}}$)**: Rechecks dynamic consent $\Sigma(t_k^{(2)})$, policy $\Pi(t_k^{(2)})$, role/purpose permissions (a, r, p) , and deterministic clinical invariants (e.g., drug-drug interactions and dosage bounds).

IV. BENCHMARK DESIGN AND SYSTEMS OPTIMIZATION

A. System Arms and Boundary Coverage

GovRed-Health evaluates four isolated research configurations:

- **UNBOUND**: LLM proposals write back directly without snapshot or governance binding (violates $\mathcal{B}_{\text{fresh}}, \mathcal{B}_{\text{causal}}, \mathcal{B}_{\text{scope}}, \mathcal{B}_{\text{admiss}}$).
- **STATE_VERSION_ONLY**: Validates base state version $\vec{v}_s$ but ignores snapshot digest, consent, and policy changes (enforces $\mathcal{B}_{\text{causal}}$ only).
- **SNAPSHOT_BOUND_STATE_ONLY**: Binds proposal to snapshot ID/digest and state version, but omits commit-time consent/policy revalidation (enforces $\mathcal{B}_{\text{causal}}, \mathcal{B}_{\text{scope}}$, fails $\mathcal{B}_{\text{fresh}}, \mathcal{B}_{\text{admiss}}$).
- **GLHS_STRICT**: Fully enforces all four Santos-Grueiro boundaries ($\mathcal{B}_{\text{fresh}} \wedge \mathcal{B}_{\text{causal}} \wedge \mathcal{B}_{\text{scope}} \wedge \mathcal{B}_{\text{admiss}}$) via a Governed State Transition (GST) gate.

B. KV-Cache Optimization and Token Minimization

A major challenge in stateful clinical AI is context bloat. Monolithic prompt architectures ingest entire patient histories, causing severe KV-cache memory pressure, cache thrashing, and high Time-To-First-Token (TTFT) latency during multi-turn agent deliberation.

GovRed-Health integrates Task-Bounded Health State Snapshot (THSS) compilation at Layer 1 (Deterministic Reconciliation Kernel). THSS applies deterministic temporal filtering ($\tau_v \cap \tau_k$) and purpose-driven entity projection, reducing prompt size by **87.4%** (mean 412 tokens vs. 3,280

TABLE III
SEALED FOUR-ARM HEADLINE RESULTS ON PRIMARY SET ($n = 210$). WILSON 95% CIs ARE REPORTED. ALL 30 STRICT RESIDUALS ARE INDETERMINATE CONCURRENT RACES, NOT CONFIRMED INVALID COMMITS.

Arm	Non-Safe Composite	Rate (95% CI)	Proportion
UNBOUND	120/210	0.571 (0.504–0.636)	
STATE_VERSION_ONLY	90/210	0.429 (0.364–0.496)	
SNAPSHOT_BOUND_STATE	90/210	0.429 (0.364–0.496)	
GLHS_STRICT	30/210	0.143 (0.102–0.197)	

tokens for full history, $p < 10^{-40}$) and inference latency by **68.2%**.

Because THSS structures context as a deterministic, canonically sorted Merkle prefix, intermediate KV-cache states are shared across multi-turn reasoning steps and verification subagents. This eliminates KV-cache eviction across concurrent sessions while strictly satisfying HIPAA/GDPR data minimization by omitting unneeded sensitive Protected Health Information (PHI).

C. Real-Boundary Execution and Deterministic Oracles

All benchmark schedules execute through the authentic deployed stack: Test Client → HTTP API Gateway → RBAC/Consent Middleware → Redis Cache → PostgreSQL 16 Ledger → Immutable Audit Store. Synthetic records contain high-entropy sentinel tokens to detect unauthorized disclosure. Deterministic oracles verify: (1) sentinel leakage, (2) wrong-subject exposure, (3) stale-state commit, (4) revoked-consent commit, (5) digest/expiry violation, and (6) audit lineage reconstruction.

V. ANALYSIS PLAN

The primary endpoints are: (1) Prohibited Disclosure Rate, (2) Stale/Unauthorized Commit Acceptance Rate, and (3) a Prespecified Non-Safe Composite. Outcomes for concurrent schedules are classified into three mutually exclusive categories: *confirmed invalid commit*, *confirmed safe rejection*, or *indeterminate ordering*. Indeterminate races are reported separately to prevent conflating unresolvable interleavings with safety or violation.

Rates are reported with 95% Wilson score confidence intervals. Matched binary comparisons use exact two-sided McNemar tests with Holm adjustment across hypothesis families. Secondary metrics include KV-cache memory footprint, prompt token counts, and end-to-end commit latency.

VI. RESULTS

A. Sealed Benchmark Execution

The sealed headline run (2026-08-17-rivf-final-003, source revision 5b2c0dbf17e2) evaluated 270 synthetic logical schedules per arm. Of these, 210 belonged to the mandatory executable primary set, and 60 were control/secondary schedules (180 protocol-excluded rows remained NOT RUN and were excluded from denominators).

TABLE IV
SYSTEMS EFFICIENCY AND CONTEXT OPTIMIZATION (STRICT THSS
VS. FULL HISTORY BASELINE, $N = 384$ COHORT).

Metric	Full History	Strict THSS	Optimization Delta
Mean Prompt Tokens	3,280	412	-87.4% ($p < 10^{-40}$)
Inference Latency (p50)	1,420 ms	452 ms	-68.2%
KV-Cache Memory / Req	128.4 MB	16.2 MB	-87.4%
PHI Over-disclosure	48.2%	0.0%	Zero PHI Leak
TOCTOU Vulnerability	57.1%	0.0%	100% Serial Fix

Under the frozen non-safe composite (Table III), GLHS_STRICT reduced non-safe outcomes from 57.1% (Unbound) to 14.3% ($b = 90, c = 0$, absolute risk reduction 42.9 percentage points, exact McNemar $p = 1.62 \times 10^{-27}$). Crucially, all 30 Strict residual cases belonged to the `concurrent_stale_state_write` family with indeterminate database serialization order. In serial authorization-drift families (consent revocation, policy drift, role mutation, subject migration, digest tampering, cache replay), GLHS_STRICT achieved **0/210 confirmed invalid commits**. Prohibited disclosure was 0/270 across all arms.

B. Token Reduction and KV-Cache Performance

Table IV summarizes the systems performance gains achieved by THSS minimization. In a 384-subject synthetic evaluation, task-bounded context compilation reduced mean prompt token consumption by **87.4%** (412 vs. 3,280 tokens, $p < 10^{-40}$) and cut p50 inference latency by **68.2%**. KV-cache allocation dropped proportionally from 128.4 MB to 16.2 MB per request, enabling higher concurrency without GPU memory thrashing while preserving 100% diagnostic accuracy ($p_{\text{TOST}} = 9.44 \times 10^{-8}$).

VII. DISCUSSION

The findings demonstrate why endpoint-level success metrics can be deceptive in stateful healthcare AI. An agent’s persistent write request may return HTTP 200 OK even though the medical evidence, dynamic consent, or clinical policy that licensed the action has drifted.

Comparing the ablation arms highlights the necessity of the Santos-Grueiro 4-boundary model. Checking state version alone ($\mathcal{B}_{\text{causal}}$) eliminates pure data overwrite races but remains blind to revoked consent or policy changes ($\mathcal{B}_{\text{admiss}}$). Binding snapshot IDs without revalidation fails when tokens are replayed across cache boundaries ($\mathcal{B}_{\text{fresh}}$). Only full four-boundary revalidation in GLHS_STRICT prevents authorization drift across the read-to-write interval. Furthermore, the 87.4% token reduction and KV-cache optimization demonstrate that rigorous governance and high computational efficiency are mutually reinforcing rather than conflicting goals.

VIII. LIMITATIONS AND ETHICS

The benchmark was conducted on synthetic clinical cohorts and a single microservice architecture; results may vary across commercial EHR databases or different policy

engines. Sentinel oracles test observable boundary integrity but do not model all semantic clinical harms. The research ablations are experimental baselines and not recommended for clinical production.

IX. REPRODUCIBILITY AND ARTIFACT CONTRACT
The complete benchmark suite is sealed under Git commit `5b2c0dbf17e2` with SHA-256 manifest hashes, frozen database migrations, Docker images, and raw JSON execution logs. Every table and statistic in this paper is directly reproducible from the sealed run artifacts.

X. CONCLUSION

GovRed-Health establishes a reproducible, real-boundary benchmark for bitemporal authorization drift in stateful clinical AI agents. By integrating the Santos-Grueiro four-boundary commit-time authorization framework with task-bounded context minimization, the benchmark demonstrates that full 4-boundary enforcement eliminates 100% of confirmed serial authorization-drift failures while delivering an 87.4% prompt token reduction and KV-cache optimization.

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